

Cloud ML Platforms: AWS vs Azure vs GCP (Vertex AI, SageMaker, Azure ML)

Big picture

All three clouds cover the **end-to-end ML lifecycle** (data → train → tune → deploy → monitor → govern). The names differ, but capabilities line up closely. Use the matrix below as your mental map; it's straight from your comparison slides.

1) Compute for Training & Inference

- **AWS:** EC2 (VMs), **ECS** (Docker service), **EKS** (managed K8s), **Fargate** (serverless pods/containers).
- **Azure:** ND A100 v4 GPU VMs, **Azure ML Compute Clusters**, Batch AI.
- **GCP:** A2/A3 GPU VMs, **Vertex AI Training**, **Cloud TPU v5e** (TensorFlow/TPU).

When to pick what (quick cues)

- You need raw control & custom AMIs → EC2 / Azure VMs / GCP A2.
 - You want managed training → Vertex AI Training / Azure ML jobs / SageMaker Training.
 - You're doing TF on TPUs → GCP Cloud TPU v5e.
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2) Containers & Serverless Compute

- **AWS:** EKS (K8s), ECS Fargate, Lambda.
- **Azure:** AKS, Container Apps, Azure Functions.

- **GCP: GKE Autopilot, Cloud Run, Cloud Functions.**

Tip: “Managed K8s” (EKS/AKS/GKE) for long-running microservices; “Functions/Cloud Run/Lambda” for quick endpoints/cron-like jobs.

3) Storage & Data Lakes

- **AWS: S3, Lake Formation, EFS/FSx.**
 - **Azure: Blob Storage, Data Lake Storage Gen2, Azure Files.**
 - **GCP: Cloud Storage, BigLake, Filestore.**
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4) Data Engineering Pipelines (ETL/ELT)

- **AWS: Glue, Data Pipeline, DMS, Kinesis.**
 - **Azure: Data Factory, Synapse Pipelines, Event Hubs.**
 - **GCP: Dataflow, Dataproc, Pub/Sub.**
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5) Feature & Embedding Stores

- **AWS: SageMaker Feature Store.**
 - **Azure: Azure ML Feature Store (Preview).**
 - **GCP: Vertex AI Feature Store.**
Purpose: consistent features for **training & serving**; solve **train–serve skew**; enable offline (batch) & online (low-latency) access.
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6) Experiment Tracking & Hyperparameter Tuning

- **AWS:** SageMaker Experiments, HPO Jobs.
 - **Azure:** ML Experiments, Sweep Jobs, **NNI** integration.
 - **GCP:** Vertex AI Experiments, **Vizier** optimizer.
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7) Deployment & Serving

- **AWS:** SageMaker Real-Time/Async Endpoints, JumpStart model zoo, **Bedrock** for LLMs.
 - **Azure:** Managed Online Endpoints, Azure OpenAI Service, K8s inference.
 - **GCP:** Vertex AI Endpoints, Generative AI platform (Gemini/Imagen).
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8) Monitoring & Observability

- **AWS:** SageMaker Model Monitor, CloudWatch, Bedrock Guardrails.
 - **Azure:** Data Drift Monitor, Application Insights, Prompt Flow evals.
 - **GCP:** Vertex Model Monitoring, Cloud Monitoring, AIP evaluation.
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9) Security & Access Control

- **AWS:** IAM, KMS, PrivateLink.
- **Azure:** RBAC, Key Vault, Private Link, Managed Identities.
- **GCP:** IAM, Cloud KMS, VPC Service Controls.

10) LLM-Specific & RAG Tooling

- **AWS:** Bedrock Knowledge Bases, Kendra, SageMaker RAG templates.
- **Azure:** Azure AI Search, Prompt Flow, LangChain on Azure.
- **GCP:** Vertex AI Search & Conversation, RAG Toolkit, grounding evals.

11) Notable Advanced Offerings

- **AWS:** CodeWhisperer, Inferentia/Trainium accelerators, SageMaker Clarify.
- **Azure:** DeepSpeed Inference, ONNX Runtime-Triton, AI Studio evals.
- **GCP:** Cloud TPU Pods, GenAI Studio, Data Fusion DQ.

12) End-to-End Pattern to Memorize

Ingestion → Feature Store → Training → Registry → CI/CD → Endpoint.

Rollout patterns for safe releases: **Blue/Green, Shadow, Canary** (esp. for LLM prompt changes).

Exam tip: know one concrete mapping per cloud from each row (compute, store, pipeline, deploy, monitor).

MCQs (15) — with answers & short explanations

1. Which GCP service is purpose-built for model training with hardware accelerators (incl. TPUs)?

A. Cloud Functions B. Vertex AI Training C. Dataflow D. Cloud Run

Answer: B. Managed training; TPUs via Cloud TPU v5e integrate tightly.

2. **On AWS, which managed K8s service would you pick for containerized model serving?**

A. ECS B. EKS C. Lambda D. Glue

Answer: B. EKS = managed Kubernetes.

3. **Which Azure option serves as the managed Kubernetes platform?**

A. AKS B. Container Apps C. Functions D. App Insights

Answer: A. AKS is Azure Kubernetes Service.

4. **Pick the correct “feature store” on GCP:**

A. BigQuery B. Vertex AI Feature Store C. Cloud Storage D. Pub/Sub

Answer: B. Vertex AI Feature Store.

5. **Which AWS service is used for real-time endpoint deployment of ML models?**

A. Lambda@Edge B. SageMaker Real-Time Endpoints C. S3 Select D. CloudTrail

Answer: B. Real-time/async endpoints live in SageMaker.

6. **You need serverless containers without managing nodes on AWS. Choose:**

A. EKS B. ECS on EC2 C. Fargate D. CloudWatch

Answer: C. Fargate = serverless for containers.

7. **Which GCP service runs containerized HTTP services with autoscale, per-request billing?**

A. Cloud Run B. GKE Autopilot C. Cloud Functions D. Dataproc

Answer: A. Cloud Run = serverless containers.

8. **Azure’s built-in drift detection for tabular ML is surfaced via:**

A. Key Vault B. Data Drift Monitor C. Kusto D. Event Hubs

Answer: B. Data Drift Monitor.

9. **Which matches “experiment tracking + HPO” on GCP?**

A. Experiment Studio B. Vertex AI Experiments + Vizier C. MLflow only D. AutoML Tables only

Answer: B. As per slides.

10. **LLM platform on AWS providing model access and guardrails:**

A. Bedrock B. JumpStart C. CodeWhisperer D. Polly

Answer: A. Bedrock + guardrails surfaced in monitoring row.

11. **Which pairing is correct for “observability on AWS”?**

A. CloudWatch + Model Monitor B. App Insights + Key Vault C. Cloud Monitoring +

Vizier

Answer: A. AWS stack.

12. **Security secrets manager on Azure used by ML workloads:**

A. KMS B. Key Vault C. IAM D. VPC SC

Answer: B. Key Vault.

13. **You need TPU-based training for TF models:**

A. EKS B. Cloud TPU v5e C. AKS D. ECS

Answer: B. GCP TPUs.

14. **Pick the LLM/RAG tool on Azure:**

A. Bedrock Knowledge Bases B. Vertex AI Search & Conversation C. Azure AI Search + Prompt Flow

Answer: C. As listed in slides.

15. **A safe rollout pattern for model updates that mirrors live traffic without user impact:**

A. Shadow B. Full cut-over C. Blue D. Rolling restart only

Answer: A. Shadow/Blue-Green/Canary patterns highlighted in the reference architecture.

Short-Answer Questions (10) — 2–3 sentences each

1. **Contrast EKS, AKS, and GKE Autopilot for ML serving.**

They're all managed Kubernetes; you package the model in a container and deploy as pods. GKE Autopilot and Fargate reduce node management; AKS integrates tightly with Azure ML endpoints.

2. **When would you choose serverless over K8s for inference?**

For spiky, unpredictable traffic and simple HTTP inference (short-lived), serverless (Cloud Run/Lambda/Functions) cuts ops overhead and bills per request. Long-running complex services usually go to K8s.

3. **Why do feature stores matter in production?**

They give consistent features for train & serve, avoiding train-serve skew; you get offline (batch) + online (low-latency) stores and lineage/ownership.

4. **Name an HPO/experiment stack on GCP and its AWS analog.**
GCP: Vertex AI Experiments + Vizier. AWS: SageMaker Experiments + HPO Jobs.
 5. **Map a minimal pipeline on AWS from data to serving.**
S3/Lake Formation → Glue ETL → SageMaker Training/Experiments → Model Monitor + Real-Time Endpoints on SageMaker.
 6. **How do you monitor a production endpoint on Azure?**
Use Data Drift Monitor for ML data quality and Application Insights for app metrics/telemetry; wire alerts into DevOps.
 7. **What makes GCP attractive for deep learning?**
Native TPUs (Cloud TPU v5e/TPU Pods), Vertex AI Training, and tight TF integration — good for high-throughput DL training.
 8. **Give an example of LLM/RAG tooling per cloud.**
AWS: Bedrock Knowledge Bases + Kendra; Azure: AI Search + Prompt Flow; GCP: Vertex AI Search & Conversation + RAG Toolkit.
 9. **What security components show up again and again across clouds?**
IAM/RBAC for authZ, managed key stores (KMS/Key Vault/Cloud KMS), and private networking (PrivateLink/VPC SC).
 10. **Describe one safe deployment pattern you'd adopt for a new model.**
Use **Canary** to send 5–10% traffic to the new model; or **Shadow** to mirror traffic without impacting users; or **Blue/Green** for instant rollback.
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Tiny “fast recap”

- Memorize one mapping per row (Compute, Containers/Serverless, Storage, Pipelines, Feature Store, Experiments/HPO, Endpoints, Monitoring, Security, LLM/RAG).
- For spiky inference → **serverless containers**; for steady high-throughput microservices → **managed K8s**.
- All three clouds have **feature stores**, **experiments/HPO**, **endpoints**, and **monitoring** with different names.

CLOUD MACHINE LEARNING TOOLS (Vertex AI, SageMaker, Azure ML)

1. Why Cloud ML Tools Exist

Machine Learning projects require **large compute, massive datasets, scalability, versioning, and continuous retraining**.

Running these on a laptop or on-premises servers is inefficient.

Cloud ML tools provide:

- **End-to-end automation** (data → training → deployment → monitoring)
- **Scalability** (scale GPUs/TPUs up or down anytime)
- **Integration** with DevOps pipelines (CI/CD, tracking)
- **Collaboration** (multiple teams share datasets, features, and models)
- **Pay-per-use** model — no upfront infrastructure cost.

So they make **MLOps** practical.

2. Common Features Across All Platforms

Stage	Cloud Tool Functions
Data Handling	Import, clean, and preprocess data at scale (Dataflow, Glue, Synapse, etc.)
Feature Store	Central location to store and serve features for ML models

Experiment Tracking	Track metrics, parameters, and model versions
Model Registry	Store trained models with version control
Deployment	Deploy as REST APIs or batch jobs
Monitoring	Detect drift, anomalies, or bias in production
Security	IAM (Identity & Access Management), encryption, audit logs

All 3 platforms (AWS, Azure, GCP) aim to simplify this **ML lifecycle** — only their ecosystems differ.

3. Major Cloud ML Platforms

A. AWS SAGEMAKER

Amazon's all-in-one MLOps platform.

Key features:

- **SageMaker Studio** → browser-based IDE (train, tune, deploy models)
- **SageMaker JumpStart** → pre-built templates & models
- **SageMaker Feature Store** → centralized features
- **SageMaker Pipelines** → automated CI/CD for ML
- **SageMaker Model Monitor** → tracks drift and bias
- **SageMaker Inference Endpoints** → deploy real-time APIs

- **AWS Bedrock** → specialized for LLMs (Anthropic, Amazon Titan, etc.)
- **Infrastructure:** EC2 (VMs), ECS/EKS (containers), Fargate (serverless pods)

Best for: Enterprises already using AWS ecosystem.

◆ B. MICROSOFT AZURE MACHINE LEARNING

Microsoft's managed ML platform tightly integrated with Azure services.

Key features:

- **Azure ML Studio** → GUI + code-based interface for ML workflow
- **Azure AutoML** → automates preprocessing, tuning, and model selection
- **Azure ML Pipelines** → reusable workflows for model training and testing
- **Azure Feature Store (Preview)** → stores features across teams
- **Managed Endpoints** → deploy models via REST APIs or containers
- **Data Drift Monitor** → tracks changes in input data
- **Azure OpenAI Service** → integrates GPT models for LLMOps
- **Integration:** Works with Power BI, Synapse Analytics, and GitHub Actions

Best for: Organizations already using Microsoft services (Office, Azure AD, etc.).

▲ C. GOOGLE CLOUD VERTEX AI

Google's modern and developer-friendly MLOps platform.

Key features:

- **Vertex AI Workbench** → unified Jupyter + BigQuery + pipelines
- **Vertex AI Pipelines** → Kubeflow-based ML workflows

- **Vertex AI Feature Store** → online + offline feature serving
- **Vertex AI Experiments** → logs metrics and training runs
- **Vertex AI Model Registry** → model storage and versioning
- **Vertex AI Endpoints** → deploy and serve models
- **Vertex AI Search & Conversation** → tools for LLMs and RAG
- **Hardware:** Supports TPUs (Tensor Processing Units)
- **Integrations:** BigQuery, Cloud Storage, Dataflow, Looker

Best for: TensorFlow users or teams preferring Google’s AI ecosystem.

 4. Cloud Comparison Summary

Feature	AWS Sagemaker	Azure ML	GCP Vertex AI
Compute	EC2, ECS, EKS, Fargate	NDv4 VMs, Batch AI	TPU v5e, A2/A3 VMs
Storage	S3, EFS, FSx	Blob, Data Lake Gen2	Cloud Storage, BigLake
ETL Tools	Glue, Data Pipeline	Data Factory, Synapse	Dataflow, Dataproc
Feature Store	SageMaker Feature Store	ML Feature Store	Vertex AI Feature Store

Experiment Tracking	SageMaker Experiments	Azure ML Experiments	Vertex Experiments
Deployment	SageMaker Endpoints	Managed Endpoints	Vertex AI Endpoints
Monitoring	CloudWatch, Model Monitor	Data Drift Monitor	Vertex Monitoring
LLM Tools	Bedrock, Kendra	Azure OpenAI, Prompt Flow	Vertex AI Search, RAG Toolkit
IaC	CloudFormation	ARM Templates	Deployment Manager
Security	IAM, KMS	RBAC, Key Vault	IAM, KMS

5. Typical End-to-End ML Pipeline in Cloud

1 Data Ingestion:

Upload to S3 / Blob / Cloud Storage

2 Feature Store:

Prepare, store, reuse data features

3 Training:

Run training job on GPU/TPU

4 Model Registry:

Save model with version info

5 Deployment:

Push to an endpoint (REST API)

6 Monitoring:

Track drift, latency, bias

7 Retraining:

Trigger retraining pipelines on drift detection

This automation = true **MLOps** lifecycle.

6. Security & Cost Management

- All clouds support **role-based access (IAM)** and **encryption (KMS, Key Vault)**.
- **PrivateLink/VPCs** isolate workloads.
- **Cost control:** Spot instances (AWS), low-priority VMs (Azure), preemptible VMs (GCP).

7. LLMOps Extensions

Modern MLOps → **LLMOps** (Large Language Model Operations).

Each platform supports:

- Prompt engineering
- Evaluation pipelines (Prompt Flow, Bedrock Guardrails, Grounding Evaluations)
- RAG (Retrieval-Augmented Generation)
- Model fine-tuning and monitoring

MCQs (15)

- 1 AWS service for feature storage → **SageMaker Feature Store**
- 2 Azure's managed ML service → **Azure Machine Learning**
- 3 GCP's ML lifecycle platform → **Vertex AI**
- 4 SageMaker component for visual interface → **SageMaker Studio**
- 5 Vertex AI's hardware accelerator → **TPU (Tensor Processing Unit)**
- 6 Azure's AutoML feature automates → **Model selection and tuning**
- 7 AWS service for real-time endpoints → **SageMaker Endpoints**
- 8 Azure's monitoring service for drift → **Data Drift Monitor**
- 9 GCP's optimizer for hyperparameter tuning → **Vertex Vizier**
- 10 LLM platform by AWS → **Bedrock**
- 11 Azure tool for LLM evaluation → **Prompt Flow**

- 12 Google's tool for conversational LLMs → **Vertex AI Search & Conversation**
 - 13 Tool to define cloud infrastructure as code in AWS → **CloudFormation**
 - 14 Tool for secure key management in Azure → **Key Vault**
 - 15 Common goal of all Cloud ML tools → **Automate the ML lifecycle**
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SHORT ANSWER QUESTIONS (10)

1 **What is Vertex AI?**

Google's unified MLOps platform combining training, deployment, and monitoring in one place.

2 **Define SageMaker.**

AWS service offering a complete environment for data scientists to train, deploy, and manage models.

3 **What is Azure ML Studio?**

A drag-and-drop or code-based workspace to design, test, and deploy ML models.

4 **What is a Feature Store?**

Centralized storage that keeps consistent, reusable data features for both training and serving models.

5 **Difference between AutoML and SageMaker.**

AutoML automates model building; SageMaker gives full control + automation over custom models.

6 **Why is drift detection important?**

To detect when real-world data distribution changes from training data, signaling model degradation.

7 **What is Vertex AI Vizier used for?**

Hyperparameter optimization (automated tuning) for model experiments.

8 **What is Bedrock?**

AWS's platform for deploying and customizing large language models securely.

9 **What are managed endpoints?**

Ready-to-use deployment endpoints where models are exposed as REST APIs.

10 **What are the advantages of cloud-based ML?**

Scalability, collaboration, automation, security, and reduced cost.